1. evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} \frac{dxdydz}{\sqrt{1-x^2-y^2-z^2}}$ by changing to spherical coordinates. Ans: $\frac{\pi^2}{8}$

2. Evaluate $\iiint_V \frac{dxdydz}{\sqrt{x^2+y^2+z^2}}$ where V is the volume of the sphere $x^2+y^2+z^2=a^2$ ans: $2\pi a^2$

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